

SOLIDWORKS API (.NET VBA C#) — Reference

(180/337): IModelDoc2 (part 3 of 17)

SOLIDWORKS API Documentation (converted from CHM)

IModelDoc2 (part 3 of 17) (Type)

SOLIDWORKS API Help

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Create3PointArc Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : Create3PointArc Method (IModelDoc2)

Obsolete. Superseded by [ISketchManager::Create3PointArc](#).

.NET Syntax

Visual Basic (Declaration)

```
Function Create3PointArc( _  
    ByVal P1x As System.Double, _  
    ByVal P1y As System.Double, _  
    ByVal P1z As System.Double, _  
    ByVal P2x As System.Double, _  
    ByVal P2y As System.Double, _  
    ByVal P2z As System.Double, _  
    ByVal P3x As System.Double, _  
    ByVal P3y As System.Double, _  
    ByVal P3z As System.Double _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim P1x As System.Double  
Dim P1y As System.Double  
Dim P1z As System.Double  
Dim P2x As System.Double  
Dim P2y As System.Double  
Dim P2z As System.Double  
Dim P3x As System.Double  
Dim P3y As System.Double  
Dim P3z As System.Double  
Dim value As System.Boolean  
  
value = instance.Create3PointArc(P1x, P1y, P1z, P2x, P2y, P2z, P3x, P3y, P3z)
```

C#

```
System.bool Create3PointArc(  
    System.double P1x,  
    System.double P1y,  
    System.double P1z,  
    System.double P2x,  
    System.double P2y,  
    System.double P2z,  
    System.double P3x,  
    System.double P3y,  
    System.double P3z  
)
```

C++/CLI

```
System.bool Create3PointArc(  
& System.double P1x,  
& System.double P1y,  
& System.double P1z,  
& System.double P2x,  
& System.double P2y,  
& System.double P2z,  
& System.double P3x,  
& System.double P3y,
```

```
& System.double P3z  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

P1x

X value of point 1

P1y

Y value of point 1

P1z

Z value of point 1

P2x

X value of point 2

P2y

Y value of point 2

P2z

Z value of point 2

P3x

X value of point 3

P3y

Y value of point 3

P3z

Z value of point 3

Return Value

True if successfully created, false otherwise

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::Create3PointArc](#).

See Also

[IModelDoc2 Interface](#)

[IModelDoc2 Members](#)

[IModelDoc2::CreateArc2 Method](#)

[IModelDoc2::CreateArcByCenter Method](#)

[IModelDoc2::CreateTangentArc2 Method](#)

[IModelDoc2::ICreateArc2 Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

CreateArc Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : CreateArc Method (IModelDoc2)

Obsolete. Superseded by [IModelDoc2::CreateArc2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function CreateArc( _  
    ByVal P1 As System.Object, _  
    ByVal P2 As System.Object, _  
    ByVal P3 As System.Object, _  
    ByVal Dir As System.Short _  
) As System.Boolean
```

Visual Basic (Usage)

```
Dim instance As IModelDoc2  
Dim P1 As System.Object  
Dim P2 As System.Object
```



Copy Code

```
Dim P3 As System.Object
Dim Dir As System.Short
Dim value As System.Boolean

value = instance.CreateArc(P1, P2, P3, Dir)
```

C#

```
System.bool CreateArc(
    System.object P1,
    System.object P2,
    System.object P3,
    System.short Dir
)
```

C++/CLI

```
System.bool CreateArc(
    & System.Object^ P1,
    & System.Object^ P2,
    & System.Object^ P3,
    & System.short Dir
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

P1

P2

P3

Dir

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::CreateArc](#).

See Also

[IModelDoc2 Interface](#)

[IModelDoc2 Members](#)

SOLIDWORKS API Help

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CreateArc2 Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : CreateArc2 Method (IModelDoc2)

Obsolete. Superseded by [ISketchManager::CreateArc](#).

.NET Syntax

Visual Basic (Declaration)

```
Function CreateArc2( _
    ByVal XC As System.Double, _
    ByVal YC As System.Double, _
    ByVal Zc As System.Double, _
    ByVal Xp1 As System.Double, _
    ByVal Yp1 As System.Double, _
    ByVal Zp1 As System.Double, _
    ByVal Xp2 As System.Double, _
    ByVal Yp2 As System.Double, _
    ByVal Zp2 As System.Double, _
    ByVal Direction As System.Short _
) As System.Object
```

Visual Basic (Usage)



```

Dim instance As IModelDoc2
Dim XC As System.Double
Dim YC As System.Double
Dim Zc As System.Double
Dim Xp1 As System.Double
Dim Yp1 As System.Double
Dim Zp1 As System.Double
Dim Xp2 As System.Double
Dim Yp2 As System.Double
Dim Zp2 As System.Double
Dim Direction As System.Short
Dim value As System.Object

value = instance.CreateArc2(XC, YC, Zc, Xp1, Yp1, Zp1, Xp2, Yp2, Zp2, Direction)

```

C#

```

System.object CreateArc2(
    System.double XC,
    System.double YC,
    System.double Zc,
    System.double Xp1,
    System.double Yp1,
    System.double Zp1,
    System.double Xp2,
    System.double Yp2,
    System.double Zp2,
    System.short Direction
)

```

C++/CLI

```

System.Object^ CreateArc2(
    & System.double XC,
    & System.double YC,
    & System.double Zc,
    & System.double Xp1,
    & System.double Yp1,
    & System.double Zp1,
    & System.double Xp2,
    & System.double Yp2,
    & System.double Zp2,
    & System.short Direction
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

XC
X value of the circle center point in meters

YC
Y value of the circle center point in meters

Zc
Z value of the circle center point in meters

Xp1
X value of the start point of the arc in meters

Yp1
Y value of the start point of the arc in meters

Zp1
Z value of the start point of the arc in meters

Xp2
X value of the end point of the arc in meters

Yp2
Y value of the end point of the arc in meters

Zp2
Z value of the end point of the arc in meters

Direction

- +1 : Go from the start point to the end point in a counter-clockwise direction

- -1 : Go from the start point to the end point in a clockwise direction

Return Value

Newly created arc (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::CreateArc2](#).

Remarks

This method creates a partial arc in the active 2D sketch. If a sketch is not active, then a new sketch is created. You can check for an active sketch using [IModelDoc2::GetActiveSketch2](#) or [IModelDoc2::IGetActiveSketch2](#).

For COM applications, the object pointer returned from this method can be used to call any APIs on the [ISketchSegment](#) interface. The underlying [ISketchArc](#) object can be obtained using QueryInterface on the returned ISketchSegment object.

OLE applications can define a new ISketchSegment or ISketchArc object using the returned Dispatch pointer. Visual Basic applications interpret the pointer for you automatically, so you can use the returned object to call ISketchSegment or ISketchArc functions.

[IModelDoc2::SetAddToDB](#) and [IModelDoc2::SetDisplayWhenAdded](#) increase performance during entity creation by adding entities directly to the SOLIDWORKS database.

- IModelDoc2::SetAddToDB also avoids some of the peculiarities involved with creating entities via the user interface, such as inferencing, automatic relations, and snapping to the grid. Adding entities directly to the database also increases the performance of this API. When you are done creating entities, it is important to call IModelDoc2::SetAddToDB(false), to restore SOLIDWORKS to its normal operating mode.
- This method also works with IModelDoc2::SetDisplayWhenAdded. If you have called IModelDoc2::SetAddToDB(True), additional performance can be gained by calling IModelDoc2::SetDisplayWhenAdded(false) to disable immediate display of entities as they are added to the database. When you are done creating all of your sketch entities, you must redraw your document window (see [IModelDoc2::GraphicsRedraw2](#)) to see the entities that you added. You should also restore the original display settings by calling IModelDoc2::SetDisplayWhenAdded(True).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

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CreateArcByCenter Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : CreateArcByCenter Method (IModelDoc2)

Creates an arc by center in this model document.

.NET Syntax

Visual Basic (Declaration)

```
Function CreateArcByCenter( _
    ByVal P1x As System.Double, _
    ByVal P1y As System.Double, _
    ByVal P1z As System.Double, _
    ByVal P2x As System.Double, _
    ByVal P2y As System.Double, _
    ByVal P2z As System.Double, _
    ByVal P3x As System.Double, _
    ByVal P3y As System.Double, _
    ByVal P3z As System.Double _
) As System.Boolean
```

Visual Basic (Usage)

```
Dim instance As IModelDoc2
Dim P1x As System.Double
Dim P1y As System.Double
Dim P1z As System.Double
Dim P2x As System.Double
Dim P2y As System.Double
```



```

Dim P2z As System.Double
Dim P3x As System.Double
Dim P3y As System.Double
Dim P3z As System.Double
Dim value As System.Boolean

```

```

value = instance.CreateArcByCenter(P1x, P1y, P1z, P2x, P2y, P2z, P3x, P3y, P3z)

```

C#

```

System.bool CreateArcByCenter(
    System.double P1x,
    System.double P1y,
    System.double P1z,
    System.double P2x,
    System.double P2y,
    System.double P2z,
    System.double P3x,
    System.double P3y,
    System.double P3z
)

```

C++/CLI

```

System.bool CreateArcByCenter(
    & System.double P1x,
    & System.double P1y,
    & System.double P1z,
    & System.double P2x,
    & System.double P2y,
    & System.double P2z,
    & System.double P3x,
    & System.double P3y,
    & System.double P3z
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

P1x
 X coordinate for first point
P1y
 Y coordinate for first point
P1z
 Z coordinate for first point
P2x
 X coordinate for second point
P2y
 Y coordinate for second point
P2z
 Z coordinate for second point
P3x
 X coordinate for third point
P3y
 Y coordinate for third point
P3z
 Z coordinate for third point

Return Value

True if successful, false otherwise

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::CreateArcByCenter](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::Create3PointArc Method](#)
[IModelDoc2::CreateArc2 Method](#)
[IModelDoc2::ICreateArc2 Method](#)
[IModelDoc2::CreateTangentArc2 Method](#)

Availability

CreateArcDB Method (IModelDoc2)

[See Also](#)[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : CreateArcDB Method (IModelDoc2)**Obsolete.** Superseded by [IModelDoc2::CreateArc2](#).

Visual Basic Syntax

Visual Basic (Declaration)

```
Function CreateArcDB( _
    ByVal X1 As System.Double, _
    ByVal Y1 As System.Double, _
    ByVal Z1 As System.Double, _
    ByVal X2 As System.Double, _
    ByVal Y2 As System.Double, _
    ByVal Z2 As System.Double, _
    ByVal X3 As System.Double, _
    ByVal Y3 As System.Double, _
    ByVal Z3 As System.Double, _
    ByVal Dir As System.Short _
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2
Dim X1 As System.Double
Dim Y1 As System.Double
Dim Z1 As System.Double
Dim X2 As System.Double
Dim Y2 As System.Double
Dim Z2 As System.Double
Dim X3 As System.Double
Dim Y3 As System.Double
Dim Z3 As System.Double
Dim Dir As System.Short
Dim value As System.Boolean
```

```
value = instance.CreateArcDB(X1, Y1, Z1, X2, Y2, Z2, X3, Y3, Z3, Dir)
```

C#

```
System.bool CreateArcDB(
    System.double X1,
    System.double Y1,
    System.double Z1,
    System.double X2,
    System.double Y2,
    System.double Z2,
    System.double X3,
    System.double Y3,
    System.double Z3,
    System.short Dir
)
```

C++/CLI

```
System.bool CreateArcDB(
    & System.double X1,
    & System.double Y1,
    & System.double Z1,
    & System.double X2,
    & System.double Y2,
    & System.double Z2,
    & System.double X3,
    & System.double Y3,
```



```
& System.double Z3,  
& System.short Dir  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

X1

Y1

Z1

X2

Y2

Z2

X3

Y3

Z3

Dir

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::CreateArcDB](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

CreateArcVB Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : CreateArcVB Method (IModelDoc2)

Obsolete. Superseded by [IModelDoc2::CreateArc2](#).

.NET Syntax

Visual Basic (Declaration)

```
Sub CreateArcVB( _  
    ByVal P1x As System.Double, _  
    ByVal P1y As System.Double, _  
    ByVal P1z As System.Double, _  
    ByVal P2x As System.Double, _  
    ByVal P2y As System.Double, _  
    ByVal P2z As System.Double, _  
    ByVal P3x As System.Double, _  
    ByVal P3y As System.Double, _  
    ByVal P3z As System.Double, _  
    ByVal Dir As System.Short _  
)
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim P1x As System.Double  
Dim P1y As System.Double  
Dim P1z As System.Double  
Dim P2x As System.Double  
Dim P2y As System.Double  
Dim P2z As System.Double  
Dim P3x As System.Double  
Dim P3y As System.Double  
Dim P3z As System.Double  
Dim Dir As System.Short
```

```
instance.CreateArcVB(P1x, P1y, P1z, P2x, P2y, P2z, P3x, P3y, P3z, Dir)
```

C#

```
void CreateArcVB(  
    System.double P1x,  
    System.double P1y,  
    System.double P1z,  
    System.double P2x,  
    System.double P2y,  
    System.double P2z,  
    System.double P3x,  
    System.double P3y,  
    System.double P3z,  
    System.short Dir  
)
```

C++/CLI

```
void CreateArcVB(  
& System.double P1x,  
& System.double P1y,  
& System.double P1z,  
& System.double P2x,  
& System.double P2y,  
& System.double P2z,  
& System.double P3x,  
& System.double P3y,  
& System.double P3z,  
& System.short Dir  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

P1x

P1y

P1z

P2x

P2y

P2z

P3x

P3y

P3z

Dir

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::CreateArcVB](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

CreateCenterLine Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : CreateCenterLine Method (IModelDoc2)

Obsolete. Superseded by [ISketchManager::CreateCenterLine](#).

.NET Syntax

Visual Basic (Declaration)

```
Function CreateCenterLine( _  
    ByVal P1 As System.Object, _  
    ByVal P2 As System.Object _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim P1 As System.Object  
Dim P2 As System.Object  
Dim value As System.Boolean  
  
value = instance.CreateCenterLine(P1, P2)
```

C#

```
System.bool CreateCenterLine(  
    System.object P1,  
    System.object P2  
)
```

C++/CLI

```
System.bool CreateCenterLine(  
& System.Object^ P1,  
& System.Object^ P2  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

P1

Array of 3 doubles (x1, y1, z1) in meters that describe the first point of the line

P2

Array of 3 doubles (x2, y2, z2) in meters that describe the second point of the line

Return Value

True if success, false if not

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::CreateCenterLine](#).

Remarks

Use [IModelDoc2::CreateCenterLineVB](#) for Visual Basic for Applications (VBA) and other forms of Basic that do not support SafeArrays.

You can also create centerline construction geometry using [IModelDoc2::CreateLine2](#) and [ISketchSegment::ConstructionGeometry](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::CreateCenterLine Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

CreateCenterLineVB Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : CreateCenterLineVB Method (IModelDoc2)

Creates a center line from P1 to P2 and can be used in Visual Basic for Applications (VBA) and other forms of Basic that do not support SafeArrays.

.NET Syntax

Visual Basic (Declaration)

```
Sub CreateCenterLineVB( _  
    ByVal X1 As System.Double, _  
    ByVal Y1 As System.Double, _  
    ByVal Z1 As System.Double, _  
    ByVal X2 As System.Double, _  
    ByVal Y2 As System.Double, _  
    ByVal Z2 As System.Double _  
)
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim X1 As System.Double  
Dim Y1 As System.Double  
Dim Z1 As System.Double  
Dim X2 As System.Double  
Dim Y2 As System.Double  
Dim Z2 As System.Double  
  
instance.CreateCenterLineVB(X1, Y1, Z1, X2, Y2, Z2)
```

C#

```
void CreateCenterLineVB(  
    System.double X1,  
    System.double Y1,  
    System.double Z1,  
    System.double X2,  
    System.double Y2,  
    System.double Z2  
)
```

C++/CLI

```
void CreateCenterLineVB(  
& System.double X1,  
& System.double Y1,  
& System.double Z1,  
& System.double X2,  
& System.double Y2,  
& System.double Z2  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

[X1](#)
Location of first end point in meters
[Y1](#)
Location of first end point in meters
[Z1](#)
Location of first end point in meters
[X2](#)
Location of second end point in meters
[Y2](#)
Location of second end point in meters
[Z2](#)
Location of second end point in meters

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::CreateCenterLineVB](#).

Remarks

You can also create centerline construction geometry using [IModelDoc2::CreateLine2](#) and [ISketchSegment::ConstructionGeometry](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::CreateCenterLine Method](#)
[IModelDoc2::ICreateCenterLine Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

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CreateCircle Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : CreateCircle Method (IModelDoc2)

Obsolete. Superseded by [IModelDoc2::CreateCircle2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function CreateCircle( _  
    ByVal P1x As System.Double, _  
    ByVal P1y As System.Double, _  
    ByVal P1z As System.Double, _  
    ByVal P2x As System.Double, _  
    ByVal P2y As System.Double, _  
    ByVal P2z As System.Double _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim P1x As System.Double  
Dim P1y As System.Double  
Dim P1z As System.Double  
Dim P2x As System.Double  
Dim P2y As System.Double  
Dim P2z As System.Double  
Dim value As System.Boolean
```

```
value = instance.CreateCircle(P1x, P1y, P1z, P2x, P2y, P2z)
```

C#

```
System.bool CreateCircle(  
    System.double P1x,  
    System.double P1y,  
    System.double P1z,  
    System.double P2x,  
    System.double P2y,  
    System.double P2z  
)
```

C++/CLI

```
System.bool CreateCircle(  
& System.double P1x,  
& System.double P1y,  
& System.double P1z,  
& System.double P2x,  
& System.double P2y,  
& System.double P2z  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

P1x

P1y

P1z

P2x

P2y

P2z

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::CreateCircle](#).

See Also

[IModelDoc2 Interface](#)

[IModelDoc2 Members](#)

SOLIDWORKS API Help

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CreateCircle2 Method (IModelDoc2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : CreateCircle2 Method (IModelDoc2)

Obsolete. Superseded by [SketchManager::CreateCircle](#).

.NET Syntax

Visual Basic (Declaration)

```
Function CreateCircle2( _  
    ByVal XC As System.Double, _  
    ByVal YC As System.Double, _  
    ByVal Zc As System.Double, _  
    ByVal Xp As System.Double, _  
    ByVal Yp As System.Double, _  
    ByVal Zp As System.Double _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim XC As System.Double  
Dim YC As System.Double  
Dim Zc As System.Double  
Dim Xp As System.Double  
Dim Yp As System.Double  
Dim Zp As System.Double  
Dim value As System.Object  
  
value = instance.CreateCircle2(XC, YC, Zc, Xp, Yp, Zp)
```

C#

```
System.object CreateCircle2(  
    System.double XC,  
    System.double YC,  
    System.double Zc,  
    System.double Xp,  
    System.double Yp,  
    System.double Zp  
)
```

C++/CLI

```
System.Object^ CreateCircle2(  
& System.double XC,  
& System.double YC,  
& System.double Zc,
```

```
& System.double Xp,  
& System.double Yp,  
& System.double Zp  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

XC

X value of the circle center point in meters

YC

Y value of the circle center point in meters

Zc

Z value of the circle center point in meters

Xp

X value of the point on the circle in meters

Yp

Y value of the point on the circle in meters

Zp

Z value of the point on the circle in meters

Return Value

Newly created circle (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::CreateCircle2](#).

Example

[Create Detail View \(VBA\)](#)

[Create Revolve Features \(VBA\)](#)

Remarks

This method creates a partial arc in the active 2D sketch. If a sketch is not active, then a new sketch is created. You can check for an active sketch using [IModelDoc2::GetActiveSketch2](#) or [IModelDoc2::IGetActiveSketch2](#).

For COM applications, the object pointer returned from this method can be used to call any APIs on the [ISketchSegment](#) interface. The underlying [ISketchArc](#) object can be obtained using QueryInterface on the returned ISketchSegment object.

OLE applications can define a new ISketchSegment or ISketchArc object using the returned Dispatch pointer. Visual Basic applications interpret the pointer for you automatically, so you can use the returned object to call ISketchSegment or ISketchArc functions.

[IModelDoc2::SetAddToDB](#) and [IModelDoc2::SetDisplayWhenAdded](#) increase performance during entity creation by adding entities directly to the SOLIDWORKS database.

- IModelDoc2::SetAddToDB also avoids some of the peculiarities involved with creating entities via the user interface, such as inferencing, automatic relations, and snapping to the grid. Adding entities directly to the database also increases the performance of this API. When you are done creating entities, it is important to call IModelDoc2::SetAddToDB(false), to restore SOLIDWORKS to its normal operating mode.
- This method also works with IModelDoc2::SetDisplayWhenAdded. If you have called IModelDoc2::SetAddToDB(True), additional performance can be gained by calling IModelDoc2::SetDisplayWhenAdded(false) to disable immediate display of entities as they are added to the database. When you are done creating all of your sketch entities, you must redraw your document window (see [IModelDoc2::GraphicsRedraw2](#)) to see the entities that you added. You should also restore the original display settings by calling IModelDoc2::SetDisplayWhenAdded(True).

To create a circle using a center point and radius, use [IModelDoc2::CreateCircleByRadius2](#) or [IModelDoc2::ICreateCircleByRadius2](#). To create a partial arc, use [IModelDoc2::CreateArc2](#) or [IModelDoc2::ICreateArc2](#).

See Also

[IModelDoc2 Interface](#)

[IModelDoc2 Members](#)

[IModelDoc2::ICreateCircle2 Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

CreateCircleByRadius Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : CreateCircleByRadius Method (IModelDoc2)

Obsolete. Superseded by [IModelDoc2::CreateCircleByRadius2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function CreateCircleByRadius( _  
    ByVal P1 As System.Object, _  
    ByVal Radius As System.Double _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim P1 As System.Object  
Dim Radius As System.Double  
Dim value As System.Boolean  
  
value = instance.CreateCircleByRadius(P1, Radius)
```

C#

```
System.bool CreateCircleByRadius(  
    System.object P1,  
    System.double Radius  
)
```

C++/CLI

```
System.bool CreateCircleByRadius(  
& System.Object^ P1,  
& System.double Radius  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

P1

Radius

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::CreateCircleByRadius](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

CreateCircleByRadius2 Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : CreateCircleByRadius2 Method (IModelDoc2)

Obsolete. Superseded by [SketchManager::CreateCircleByRadius](#).

.NET Syntax

Visual Basic (Declaration)

```
Function CreateCircleByRadius2( _  
    ByVal XC As System.Double, _  
    ByVal YC As System.Double, _  
    ByVal Zc As System.Double, _  
    ByVal Radius As System.Double _  
) As System.Object
```



```

Dim instance As IModelDoc2
Dim XC As System.Double
Dim YC As System.Double
Dim Zc As System.Double
Dim Radius As System.Double
Dim value As System.Object

value = instance.CreateCircleByRadius2(XC, YC, Zc, Radius)

```

C#

```

System.Object CreateCircleByRadius2(
    System.double XC,
    System.double YC,
    System.double Zc,
    System.double Radius
)

```

C++/CLI

```

System.Object^ CreateCircleByRadius2(
    & System.double XC,
    & System.double YC,
    & System.double Zc,
    & System.double Radius
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

XC
 X value of the circle center point in meters
YC
 Y value of the circle center point in meters
Zc
 Z value of the circle center point in meters
Radius
 Radius of the circle in meters

Return Value

Newly created circle

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::CreateCircleByRadius2](#).

Remarks

This method creates a partial arc in the active 2D sketch. If a sketch is not active, then a new sketch is created. You can check for an active sketch using [IModelDoc2::GetActiveSketch2](#) or [IModelDoc2::IGetActiveSketch2](#).

For COM applications, the object pointer returned from this method can be used to call any APIs on the [ISketchSegment](#) interface. The underlying [ISketchArc](#) object can be obtained using QueryInterface on the returned ISketchSegment object.

OLE applications can define a new ISketchSegment or ISketchArc object using the returned Dispatch pointer. Visual Basic applications interpret the pointer for you automatically, so you can use the returned object to call ISketchSegment or ISketchArc functions.

[IModelDoc2::SetAddToDB](#) and [IModelDoc2::SetDisplayWhenAdded](#) increase performance during entity creation by adding entities directly to the SOLIDWORKS database.

- IModelDoc2::SetAddToDB also avoids some of the peculiarities involved with creating entities via the user interface, such as inferencing, automatic relations, and snapping to the grid. Adding entities directly to the database also increases the performance of this API. When you are done creating entities, it is important to call IModelDoc2::SetAddToDB(false), to restore SOLIDWORKS to its normal operating mode.
- This method also works with IModelDoc2::SetDisplayWhenAdded. If you have called IModelDoc2::SetAddToDB(True), additional performance can be gained by calling IModelDoc2::SetDisplayWhenAdded(false) to disable immediate display of entities as they are added to the database. When you are done creating all of your sketch entities, you must redraw your document window (see [IModelDoc2::GraphicsRedraw2](#)) to see the entities that you added. You should also restore the original display settings by calling IModelDoc2::SetDisplayWhenAdded(True).

To create a circle using a center point and a point on the circle, see [IModelDoc2::CreateCircle2](#) or [IModelDoc2::ICreateCircle2](#). To create a partial arc, see [IModelDoc2::CreateArc2](#) or [IModelDoc2::ICreateArc2](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::CreateCircleByRadius2 Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

CreateCircleDB Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : CreateCircleDB Method (IModelDoc2)

Obsolete. Superseded by [IModelDoc2::CreateCircle2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function CreateCircleDB( _  
    ByVal Cx As System.Double, _  
    ByVal Cy As System.Double, _  
    ByVal Cz As System.Double, _  
    ByVal Radius As System.Double _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim Cx As System.Double  
Dim Cy As System.Double  
Dim Cz As System.Double  
Dim Radius As System.Double  
Dim value As System.Boolean  
  
value = instance.CreateCircleDB(Cx, Cy, Cz, Radius)
```

C#

```
System.bool CreateCircleDB(  
    System.double Cx,  
    System.double Cy,  
    System.double Cz,  
    System.double Radius  
)
```

C++/CLI

```
System.bool CreateCircleDB(  
& System.double Cx,  
& System.double Cy,  
& System.double Cz,  
& System.double Radius  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Cx

Cy

Cz

Radius

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::CreateCircleDB](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

CreateCircleVB Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : CreateCircleVB Method (IModelDoc2)

Obsolete. Superseded by [IModelDoc2::CreateCircle2](#).

.NET Syntax

Visual Basic (Declaration)

```
Sub CreateCircleVB( _  
    ByVal Plx As System.Double, _  
    ByVal Ply As System.Double, _  
    ByVal Plz As System.Double, _  
    ByVal Radius As System.Double _  
)
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim Plx As System.Double  
Dim Ply As System.Double  
Dim Plz As System.Double  
Dim Radius As System.Double  
  
instance.CreateCircleVB(Plx, Ply, Plz, Radius)
```

C#

```
void CreateCircleVB(  
    System.double Plx,  
    System.double Ply,  
    System.double Plz,  
    System.double Radius  
)
```

C++/CLI

```
void CreateCircleVB(  
& System.double Plx,  
& System.double Ply,  
& System.double Plz,  
& System.double Radius  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Plx

Ply

Plz

Radius

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::CreateCircleVB](#).

See Also

CreateCircularSketchStepAndRepeat Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : CreateCircularSketchStepAndRepeat Method (IModelDoc2)

Obsolete. Superseded by [ISketchManager::CreateCircularSketchStepAndRepeat](#).

Visual Basic Syntax

Visual Basic (Declaration)

```
Function CreateCircularSketchStepAndRepeat( _  
    ByVal ArcRadius As System.Double, _  
    ByVal ArcAngle As System.Double, _  
    ByVal PatternNum As System.Integer, _  
    ByVal PatternSpacing As System.Double, _  
    ByVal PatternRotate As System.Boolean, _  
    ByVal DeleteInstances As System.String _  
    ) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim ArcRadius As System.Double  
Dim ArcAngle As System.Double  
Dim PatternNum As System.Integer  
Dim PatternSpacing As System.Double  
Dim PatternRotate As System.Boolean  
Dim DeleteInstances As System.String  
Dim value As System.Boolean  
  
value = instance.CreateCircularSketchStepAndRepeat(ArcRadius, ArcAngle, PatternNum,  
PatternSpacing, PatternRotate, DeleteInstances)
```

C#

```
System.bool CreateCircularSketchStepAndRepeat(  
    System.double ArcRadius,  
    System.double ArcAngle,  
    System.int PatternNum,  
    System.double PatternSpacing,  
    System.bool PatternRotate,  
    System.string DeleteInstances  
)
```

C++/CLI

```
System.bool CreateCircularSketchStepAndRepeat(  
& System.double ArcRadius,  
& System.double ArcAngle,  
& System.int PatternNum,  
& System.double PatternSpacing,  
& System.bool PatternRotate,  
& System.String^ DeleteInstances  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ArcRadius

Radius to be used in the circular sketch pattern

ArcAngle

Angle relative to the sketch entities being patterned

PatternNum

Total number of instances, including the seed geometry

PatternSpacing

Spacing between pattern instances in radians

PatternRotate

True to rotate the pattern, false to not

DeleteInstances

Number of instances to delete passed as a string formatted as: "(a) (b) (c) "

Return Value

True if the sketch pattern was created successfully, false otherwise

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::CreateCircularSketchStepAndRepeat](#).

See Also

[IModelDoc2 Interface](#)

[IModelDoc2 Members](#)

[IModelDoc2::CreateLinearSketchStepAndRepeat Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

CreateClippedSplines Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : CreateClippedSplines Method (IModelDoc2)

Creates one or more sketch spline segments that are clipped against a given (x1, y1), (x2, y2) rectangle. This rectangle lies in the space of the active 2D sketch.

.NET Syntax

Visual Basic (Declaration)

```
Function CreateClippedSplines( _  
    ByVal ParamsIn As System.Object, _  
    ByVal X1 As System.Double, _  
    ByVal Y1 As System.Double, _  
    ByVal X2 As System.Double, _  
    ByVal Y2 As System.Double _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim ParamsIn As System.Object  
Dim X1 As System.Double  
Dim Y1 As System.Double  
Dim X2 As System.Double  
Dim Y2 As System.Double  
Dim value As System.Object  
  
value = instance.CreateClippedSplines(ParamsIn, X1, Y1, X2, Y2)
```

C#

```
System.object CreateClippedSplines(  
    System.object ParamsIn,  
    System.double X1,  
    System.double Y1,  
    System.double X2,  
    System.double Y2  
)
```

C++/CLI

```
System.Object^ CreateClippedSplines(  
& System.Object^ ParamsIn,
```

```
& System.double X1,  
& System.double Y1,  
& System.double X2,  
& System.double Y2  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ParamsIn

See **Remarks**

X1

x component of the lower corner of the clipping rectangle

Y1

y component of the lower corner of the clipping rectangle

X2

x component of the upper corner of the clipping rectangle

Y2

y component of the upper corner of the clipping rectangle

Return Value

Array of newly created sketch segments

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::CreateClippedSplines](#).

Remarks

The rectangle lies in the space of the active 2D sketch. The results are undefined for calls made in an active 3D sketch.

The ParamsIn argument in Visual Basic for Applications (VBA) is a SafeArray of size (4 + numKnots + numControlPointDoubles) as follows:

[*Dimension*, *Order*, *NumControlPoints*, *Periodicity*, *knot1*, *knot2*,..., *ControlPoint1[Dimension]*, *ControlPoint2[Dimension]*,...]

where:

Dimension, *Order*, *NumControlPoints*, and *Periodicity* are integer values.

The size of the knot array is determined by (*NumControlPoints* + *Order*).

The size of the control point array is based upon the curve dimension:

- If *Dimension* = 3, then *ControlPoint* is an array of 3 doubles (*x*, *y*, *z*).
- If *Dimension* = 4, then *ControlPoint* is an array of 4 doubles (*x*, *y*, *z*, *w*) where *w* = weight.

See Also

[IModelDoc2 Interface](#)

[IModelDoc2 Members](#)

[IModelDoc2::ICreateClippedSplines Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

CreateEllipse Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : CreateEllipse Method (IModelDoc2)

Obsolete. Superseded by [IModelDoc2::CreateEllipse2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function CreateEllipse( _  
    ByVal Center As System.Object, _  
    ByVal Major As System.Object, _
```

```
ByVal Minor As System.Object _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim Center As System.Object  
Dim Major As System.Object  
Dim Minor As System.Object  
Dim value As System.Boolean  
  
value = instance.CreateEllipse(Center, Major, Minor)
```

C#

```
System.bool CreateEllipse(  
    System.object Center,  
    System.object Major,  
    System.object Minor  
)
```

C++/CLI

```
System.bool CreateEllipse(  
    & System.Object^ Center,  
    & System.Object^ Major,  
    & System.Object^ Minor  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Center

Major

Minor

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::CreateEllipse](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

CreateEllipse2 Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : CreateEllipse2 Method (IModelDoc2)

Obsolete. Superseded by [ISketchManager::CreateEllipse](#).

.NET Syntax

Visual Basic (Declaration)

```
Function CreateEllipse2( _
    ByVal CenterX As System.Double, _
    ByVal CenterY As System.Double, _
    ByVal CenterZ As System.Double, _
    ByVal MajorX As System.Double, _
    ByVal MajorY As System.Double, _
    ByVal MajorZ As System.Double, _
    ByVal MinorX As System.Double, _
    ByVal MinorY As System.Double, _
    ByVal MinorZ As System.Double _
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2
Dim CenterX As System.Double
Dim CenterY As System.Double
Dim CenterZ As System.Double
Dim MajorX As System.Double
Dim MajorY As System.Double
Dim MajorZ As System.Double
Dim MinorX As System.Double
Dim MinorY As System.Double
Dim MinorZ As System.Double
Dim value As System.Object
```

```
value = instance.CreateEllipse2(CenterX, CenterY, CenterZ, MajorX, MajorY, MajorZ, MinorX, MinorY, MinorZ)
```

C#

```
System.object CreateEllipse2(
    System.double CenterX,
    System.double CenterY,
    System.double CenterZ,
    System.double MajorX,
    System.double MajorY,
    System.double MajorZ,
    System.double MinorX,
    System.double MinorY,
    System.double MinorZ
)
```

C++/CLI

```
System.Object^ CreateEllipse2(
    & System.double CenterX,
    & System.double CenterY,
    & System.double CenterZ,
    & System.double MajorX,
    & System.double MajorY,
    & System.double MajorZ,
    & System.double MinorX,
    & System.double MinorY,
    & System.double MinorZ
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

CenterX

X values for the ellipse center point

CenterY

Y values for the ellipse center point

CenterZ

Z values for the ellipse center point

MajorX

X values for a point on the ellipse and on the major axis

MajorY

Y values for a point on the ellipse and on the major axis

MajorZ

Z values for a point on the ellipse and on the major axis

MinorX

X values for a point on the ellipse and on the minor axis
MinorY
Y values for a point on the ellipse and on the minor axis
MinorZ
Z values for a point on the ellipse and on the minor axis

Return Value

Newly created ellipse

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::CreateEllipse2](#).

Remarks

This method creates a partial arc in the active 2D sketch. If a sketch is not active, then a new sketch is created. You can check for an active sketch using [IModelDoc2::GetActiveSketch2](#) or [IModelDoc2::IGetActiveSketch2](#).

For COM applications, the object pointer returned from this method can be used to call any APIs on the [ISketchSegment](#) interface. The underlying [ISketchArc](#) object can be obtained using QueryInterface on the returned ISketchSegment object.

OLE applications can define a new ISketchSegment or ISketchArc object using the returned Dispatch pointer. Visual Basic applications interpret the pointer for you automatically, so you can use the returned object to call ISketchSegment or ISketchArc functions.

[IModelDoc2::SetAddToDB](#) and [IModelDoc2::SetDisplayWhenAdded](#) increase performance during entity creation by adding entities directly to the SOLIDWORKS database.

- IModelDoc2::SetAddToDB also avoids some of the peculiarities involved with creating entities via the user interface, such as inferencing, automatic relations, and snapping to the grid. Adding entities directly to the database also increases the performance of this API. When you are done creating entities, it is important to call IModelDoc2::SetAddToDB(false), to restore SOLIDWORKS to its normal operating mode.
- This method also works with IModelDoc2::SetDisplayWhenAdded. If you have called IModelDoc2::SetAddToDB(True), additional performance can be gained by calling IModelDoc2::SetDisplayWhenAdded(false) to disable immediate display of entities as they are added to the database. When you are done creating all of your sketch entities, you must redraw your document window (see [IModelDoc2::GraphicsRedraw2](#)) to see the entities that you added. You should also restore the original display settings by calling IModelDoc2::SetDisplayWhenAdded(True).

To create a partial ellipse, use [IModelDoc2::CreateEllipticalArcByCenter](#) or [IModelDoc2::CreateEllipticalArcByCenterVB](#) or [IModelDoc2::ICreateEllipticalArcByCenter](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::ICreateEllipse2 Method](#)
[IModelDoc2::CreateEllipticalArc2 Method](#)
[IModelDoc2::ICreateEllipticalArc2 Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

CreateEllipseVB Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : CreateEllipseVB Method (IModelDoc2)

Obsolete. Superseded by [IModelDoc2::CreateEllipse2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function CreateEllipseVB( _  
    ByVal CenterX As System.Double, _  
    ByVal CenterY As System.Double, _  
    ByVal CenterZ As System.Double, _  
    ByVal MajorX As System.Double, _  
    ByVal MajorY As System.Double, _  
    ByVal MajorZ As System.Double, _  
    ByVal MinorX As System.Double, _  
    ByVal MinorY As System.Double, _  
    ByVal MinorZ As System.Double _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim CenterX As System.Double  
Dim CenterY As System.Double  
Dim CenterZ As System.Double  
Dim MajorX As System.Double  
Dim MajorY As System.Double  
Dim MajorZ As System.Double  
Dim MinorX As System.Double  
Dim MinorY As System.Double  
Dim MinorZ As System.Double  
Dim value As System.Boolean
```

```
value = instance.CreateEllipseVB(CenterX, CenterY, CenterZ, MajorX, MajorY, MajorZ, MinorX,  
MinorY, MinorZ)
```

C#

```
System.bool CreateEllipseVB(  
    System.double CenterX,  
    System.double CenterY,  
    System.double CenterZ,  
    System.double MajorX,  
    System.double MajorY,  
    System.double MajorZ,  
    System.double MinorX,  
    System.double MinorY,  
    System.double MinorZ  
)
```

C++/CLI

```
System.bool CreateEllipseVB(  
& System.double CenterX,  
& System.double CenterY,  
& System.double CenterZ,  
& System.double MajorX,  
& System.double MajorY,  
& System.double MajorZ,  
& System.double MinorX,  
& System.double MinorY,  
& System.double MinorZ  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

CenterX

CenterY

CenterZ

MajorX

MajorY

MajorZ

MinorX

MinorY

MinorZ

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::CreateEllipseVB](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

CreateEllipticalArc2 Method (IModelDoc2)

[See Also](#)[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : CreateEllipticalArc2 Method (IModelDoc2)Obsolete. Superseded by [SketchManager::CreateEllipticalArc](#).

.NET Syntax

Visual Basic (Declaration)

```

Function CreateEllipticalArc2( _
    ByVal CenterX As System.Double, _
    ByVal CenterY As System.Double, _
    ByVal CenterZ As System.Double, _
    ByVal MajorX As System.Double, _
    ByVal MajorY As System.Double, _
    ByVal MajorZ As System.Double, _
    ByVal MinorX As System.Double, _
    ByVal MinorY As System.Double, _
    ByVal MinorZ As System.Double, _
    ByVal StartX As System.Double, _
    ByVal StartY As System.Double, _
    ByVal StartZ As System.Double, _
    ByVal EndX As System.Double, _
    ByVal EndY As System.Double, _
    ByVal EndZ As System.Double _
) As System.Object

```

Visual Basic (Usage)



```

Dim instance As IModelDoc2
Dim CenterX As System.Double
Dim CenterY As System.Double
Dim CenterZ As System.Double
Dim MajorX As System.Double
Dim MajorY As System.Double
Dim MajorZ As System.Double
Dim MinorX As System.Double
Dim MinorY As System.Double
Dim MinorZ As System.Double
Dim StartX As System.Double
Dim StartY As System.Double
Dim StartZ As System.Double
Dim EndX As System.Double
Dim EndY As System.Double
Dim EndZ As System.Double
Dim value As System.Object

```

```

value = instance.CreateEllipticalArc2(CenterX, CenterY, CenterZ, MajorX, MajorY, MajorZ, MinorX,
MinorY, MinorZ, StartX, StartY, StartZ, EndX, EndY, EndZ)

```

C#

```

System.object CreateEllipticalArc2(
    System.double CenterX,
    System.double CenterY,
    System.double CenterZ,
    System.double MajorX,
    System.double MajorY,
    System.double MajorZ,
    System.double MinorX,
    System.double MinorY,
    System.double MinorZ,
    System.double StartX,
    System.double StartY,
    System.double StartZ,
    System.double EndX,

```

```
System.double EndY,  
System.double EndZ  
)
```

C++/CLI

```
System.Object^ CreateEllipticalArc2(  
& System.double CenterX,  
& System.double CenterY,  
& System.double CenterZ,  
& System.double MajorX,  
& System.double MajorY,  
& System.double MajorZ,  
& System.double MinorX,  
& System.double MinorY,  
& System.double MinorZ,  
& System.double StartX,  
& System.double StartY,  
& System.double StartZ,  
& System.double EndX,  
& System.double EndY,  
& System.double EndZ  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

CenterX

X coordinate for the ellipse center point

CenterY

Y coordinate for the ellipse center point

CenterZ

Z coordinate for the ellipse center point

MajorX

X coordinate for a point on the ellipse and on the major axis

MajorY

Y coordinate for a point on the ellipse and on the major axis

MajorZ

Z coordinate for a point on the ellipse and on the major axis

MinorX

X coordinate for a point on the ellipse and on the minor axis

MinorY

Y coordinate for a point on the ellipse and on the minor axis

MinorZ

Z coordinate for a point on the ellipse and on the minor axis

StartX

X coordinate for counter clockwise elliptical arc start point

StartY

Y coordinate for counter clockwise elliptical arc start point

StartZ

Z coordinate for counter clockwise elliptical arc start point

EndX

X coordinate for counter clockwise elliptical arc end point

EndY

Y coordinate for counter clockwise elliptical arc end point

EndZ

Z coordinate for counter clockwise elliptical arc end point

Return Value

Newly created ellipse

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::CreateEllipticalArc2](#).

Remarks

This method creates a partial arc in the active 2D sketch. If a sketch is not active, then a new sketch is created. You can check for an active sketch using [IModelDoc2::GetActiveSketch2](#) or [IModelDoc2::IGetActiveSketch2](#).

For COM applications, the object pointer returned from this method can be used to call any APIs on the [ISketchSegment](#) interface. The underlying [ISketchArc](#) object can be obtained using QueryInterface on the returned ISketchSegment object.

OLE applications can define a new ISketchSegment or ISketchArc object using the returned Dispatch pointer. Visual Basic applications interpret the pointer for you automatically, so you can use the returned object to call ISketchSegment or ISketchArc functions.

[IModelDoc2::SetAddToDB](#) and [IModelDoc2::SetDisplayWhenAdded](#) increase performance during entity creation by adding entities directly to the SOLIDWORKS database.

- IModelDoc2::SetAddToDB also avoids some of the peculiarities involved with creating entities via the user interface, such as inferencing, automatic relations, and snapping to the grid. Adding entities directly to the database also increases the performance of this API. When you are done creating entities, it is important to call IModelDoc2::SetAddToDB(false), to restore SOLIDWORKS to its normal operating mode.
- This method also works with IModelDoc2::SetDisplayWhenAdded. If you have called IModelDoc2::SetAddToDB(True), additional performance can be gained by calling IModelDoc2::SetDisplayWhenAdded(false) to disable immediate display of entities as they are added to the database. When you are done creating all of your sketch entities, you must redraw your document window (see [IModelDoc2::GraphicsRedraw2](#)) to see the entities that you added. You should also restore the original display settings by calling IModelDoc2::SetDisplayWhenAdded(True).

To create a complete ellipse, use [IModelDoc2::CreateEllipse2](#) or [IModelDoc2::ICreateEllipse2](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::CreateEllipse2 Method](#)
[IModelDoc2::CreateEllipticalArcByCenter Method](#)
[IModelDoc2::CreateEllipticalArcByCenterVB Method](#)
[IModelDoc2::ICreateEllipse2 Method](#)
[IModelDoc2::ICreateEllipticalArc2 Method](#)
[IModelDoc2::ICreateEllipticalArcByCenter Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

CreateEllipticalArcByCenter Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : CreateEllipticalArcByCenter Method (IModelDoc2)

Obsolete. Superseded by [SketchManager::CreateEllipticalArc](#).

.NET Syntax

Visual Basic (Declaration)

```
Function CreateEllipticalArcByCenter( _  
    ByVal Center As System.Object, _  
    ByVal Major As System.Object, _  
    ByVal Minor As System.Object, _  
    ByVal Start As System.Object, _  
    ByVal End As System.Object _  
) As System.Boolean
```

Visual Basic (Usage)

```
Dim instance As IModelDoc2  
Dim Center As System.Object  
Dim Major As System.Object  
Dim Minor As System.Object  
Dim Start As System.Object  
Dim End As System.Object  
Dim value As System.Boolean
```

```
value = instance.CreateEllipticalArcByCenter(Center, Major, Minor, Start, End)
```

C#

```
System.bool CreateEllipticalArcByCenter(  
    System.object Center,  
    System.object Major,  
    System.object Minor,  
    System.object Start,
```

```
System.Object End  
)
```

C++/CLI

```
System.bool CreateEllipticalArcByCenter(  
& System.Object^ Center,  
& System.Object^ Major,  
& System.Object^ Minor,  
& System.Object^ Start,  
& System.Object^ End  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Center
Array of 3 doubles (x1, y1, z1) in meters that describe the ellipse center
Major
Array of 3 doubles (x1, y1, z1) in meters that describe a point on the ellipse and on the major axis
Minor
Array of 3 doubles (x1, y1, z1) in meters that describe a point on the ellipse and on the minor axis
Start
Array of 3 doubles (x1, y1, z1) in meters that describe the start point of the ellipse
End
Array of 3 doubles (x1, y1, z1) in meters that describe the end point of the ellipse

Return Value

True if successfully created, false otherwise

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::CreateEllipticalArcByCenter](#).

Remarks

The Start and End arguments should be specified in a counter-clockwise (CCW) manner.

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::ICreateEllipticalArcByCenter Method](#)
[IModelDoc2::ICreateEllipticalArc2 Method](#)
[IModelDoc2::ICreateEllipse2 Method](#)
[IModelDoc2::CreateEllipticalArc2 Method](#)
[IModelDoc2::CreateEllipse2 Method](#)
[IModelDoc2::CreateEllipticalArcByCenterVB Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

CreateEllipticalArcByCenterVB Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : CreateEllipticalArcByCenterVB Method (IModelDoc2)

Obsolete. Superseded by [SketchManager::CreateEllipticalArc](#).

.NET Syntax

Visual Basic (Declaration)

```
Function CreateEllipticalArcByCenterVB( _  
    ByVal CenterX As System.Double, _  
    ByVal CenterY As System.Double, _  
    ByVal CenterZ As System.Double, _  
    ByVal MajorX As System.Double, _  
    ByVal MajorY As System.Double, _  
    ByVal MajorZ As System.Double, _
```

```

ByVal MinorX As System.Double, _
ByVal MinorY As System.Double, _
ByVal MinorZ As System.Double, _
ByVal StartX As System.Double, _
ByVal StartY As System.Double, _
ByVal StartZ As System.Double, _
ByVal EndX As System.Double, _
ByVal EndY As System.Double, _
ByVal EndZ As System.Double _
) As System.Boolean

```

Visual Basic (Usage)



```

Dim instance As IModelDoc2
Dim CenterX As System.Double
Dim CenterY As System.Double
Dim CenterZ As System.Double
Dim MajorX As System.Double
Dim MajorY As System.Double
Dim MajorZ As System.Double
Dim MinorX As System.Double
Dim MinorY As System.Double
Dim MinorZ As System.Double
Dim StartX As System.Double
Dim StartY As System.Double
Dim StartZ As System.Double
Dim EndX As System.Double
Dim EndY As System.Double
Dim EndZ As System.Double
Dim value As System.Boolean

```

```

value = instance.CreateEllipticalArcByCenterVB(CenterX, CenterY, CenterZ, MajorX, MajorY, MajorZ,
MinorX, MinorY, MinorZ, StartX, StartY, StartZ, EndX, EndY, EndZ)

```

C#

```

System.bool CreateEllipticalArcByCenterVB(
    System.double CenterX,
    System.double CenterY,
    System.double CenterZ,
    System.double MajorX,
    System.double MajorY,
    System.double MajorZ,
    System.double MinorX,
    System.double MinorY,
    System.double MinorZ,
    System.double StartX,
    System.double StartY,
    System.double StartZ,
    System.double EndX,
    System.double EndY,
    System.double EndZ
)

```

C++/CLI

```

System.bool CreateEllipticalArcByCenterVB(
    & System.double CenterX,
    & System.double CenterY,
    & System.double CenterZ,
    & System.double MajorX,
    & System.double MajorY,
    & System.double MajorZ,
    & System.double MinorX,
    & System.double MinorY,
    & System.double MinorZ,
    & System.double StartX,
    & System.double StartY,
    & System.double StartZ,
    & System.double EndX,

```

```
& System.double EndY,  
& System.double EndZ  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

CenterX

X values for the ellipse center point

CenterY

Y values for the ellipse center point

CenterZ

Z values for the ellipse center point

MajorX

X values for a point on the ellipse and on the major axis

MajorY

Y values for a point on the ellipse and on the major axis

MajorZ

Z values for a point on the ellipse and on the major axis

MinorX

X values for a point on the ellipse and on the minor axis

MinorY

Y values for a point on the ellipse and on the minor axis

MinorZ

Z values for a point on the ellipse and on the minor axis

StartX

X values for CCW elliptical arc start point

StartY

Y values for CCW elliptical arc start point

StartZ

Z values for CCW elliptical arc start point

EndX

X values for CCW elliptical arc end point

EndY

Y values for CCW elliptical arc end point

EndZ

Z values for CCW elliptical arc end point

Return Value

True if successfully created, false otherwise

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::CreateEllipticalArcByCenterVB](#).

Remarks

The Start* and End* arguments should be specified in a counter-clockwise (CCW) manner.

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::ICreateEllipse2 Method](#)
[IModelDoc2::ICreateEllipticalArc2 Method](#)
[IModelDoc2::ICreateEllipticalArcByCenter Method](#)
[IModelDoc2::CreateEllipse2 Method](#)
[IModelDoc2::CreateEllipticalArc2 Method](#)
[IModelDoc2::CreateEllipticalArcByCenter Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

CreateFeatureMgrView Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : CreateFeatureMgrView Method (IModelDoc2)

Obsolete. Superseded by [IModelViewManager::CreateFeatureMgrView2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function CreateFeatureMgrView( _  
    ByRef Bitmap As System.Integer _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim Bitmap As System.Integer  
Dim value As System.Object  
  
value = instance.CreateFeatureMgrView(Bitmap)
```

C#

```
System.Object CreateFeatureMgrView(  
    ref System.int Bitmap  
)
```

C++/CLI

```
System.Object^ CreateFeatureMgrView(  
& System.int% Bitmap  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Bitmap

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::CreateFeatureMgrView](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

CreateFeatureMgrView2 Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : CreateFeatureMgrView2 Method (IModelDoc2)

Obsolete. Superseded by [IModelViewManager::CreateFeatureMgrView2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function CreateFeatureMgrView2( _  
    ByRef Bitmap As System.Integer, _  
    ByVal ToolTip As System.String _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim Bitmap As System.Integer  
Dim ToolTip As System.String  
Dim value As System.Object  
  
value = instance.CreateFeatureMgrView2(Bitmap, ToolTip)
```

C#

```
System.object CreateFeatureMgrView2(  
    ref System.int Bitmap,  
    System.string ToolTip  
)
```

C++/CLI

```
System.Object^ CreateFeatureMgrView2(  
& System.int% Bitmap,  
& System.String^ ToolTip  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Bitmap

ToolTip

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::CreateFeatureMgrView2](#).

See Also

[IModelDoc2 Interface](#)

[IModelDoc2 Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

CreateFeatureMgrView3 Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : CreateFeatureMgrView3 Method (IModelDoc2)

Obsolete. Superseded by [IModelViewManager::CreateFeatureMgrView2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function CreateFeatureMgrView3( _  
    ByRef Bitmap As System.Integer, _  
    ByVal ToolTip As System.String, _  
    ByVal WhichPane As System.Integer _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim Bitmap As System.Integer  
Dim ToolTip As System.String  
Dim WhichPane As System.Integer  
Dim value As System.Object  
  
value = instance.CreateFeatureMgrView3(Bitmap, ToolTip, WhichPane)
```

C#

```
System.object CreateFeatureMgrView3(  
    ref System.int Bitmap,  
    System.string ToolTip,  
    System.int WhichPane  
)
```

C++/CLI

```
System.Object^ CreateFeatureMgrView3(  
& System.int% Bitmap,
```

```
& System.String^ ToolTip,  
& System.int WhichPane  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Bitmap

ToolTip

WhichPane

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::CreateFeatureMgrView3](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

CreateGroup Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : CreateGroup Method (IModelDoc2)

Creates an annotation group from the currently selected annotations.

.NET Syntax

Visual Basic (Declaration)

```
Sub CreateGroup()
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
  
instance.CreateGroup()
```

C#

```
void CreateGroup()
```

C++/CLI

```
void CreateGroup();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::CreateGroup](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IAnnotation Interface](#)
[IModelDoc2::RemoveGroups Method](#)
[IModelDoc2::RemoveItemsFromGroup Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

CreateLine Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : CreateLine Method (IModelDoc2)

Obsolete. Superseded by [IModelDoc2::CreateLine2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function CreateLine(  
    ByVal P1 As System.Object, _  
    ByVal P2 As System.Object _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim P1 As System.Object  
Dim P2 As System.Object  
Dim value As System.Boolean  
  
value = instance.CreateLine(P1, P2)
```

C#

```
System.bool CreateLine(  
    System.object P1,  
    System.object P2  
)
```

C++/CLI

```
System.bool CreateLine(  
& System.Object^ P1,  
& System.Object^ P2  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

P1

P2

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::CreateLine](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

CreateLine2 Method (IModelDoc2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : CreateLine2 Method (IModelDoc2)

Obsolete. Superseded by [SketchManager::CreateLine](#).

.NET Syntax

Visual Basic (Declaration)

```
Function CreateLine2(  
    ByVal P1x As System.Double, _  
    ByVal P1y As System.Double, _
```

```

        ByVal P1z As System.Double, _
        ByVal P2x As System.Double, _
        ByVal P2y As System.Double, _
        ByVal P2z As System.Double _
    ) As System.Object

```

Visual Basic (Usage)



```

Dim instance As IModelDoc2
Dim P1x As System.Double
Dim P1y As System.Double
Dim P1z As System.Double
Dim P2x As System.Double
Dim P2y As System.Double
Dim P2z As System.Double
Dim value As System.Object

```

```
value = instance.CreateLine2(P1x, P1y, P1z, P2x, P2y, P2z)
```

C#

```

System.object CreateLine2(
    System.double P1x,
    System.double P1y,
    System.double P1z,
    System.double P2x,
    System.double P2y,
    System.double P2z
)

```

C++/CLI

```

System.Object^ CreateLine2(
    & System.double P1x,
    & System.double P1y,
    & System.double P1z,
    & System.double P2x,
    & System.double P2y,
    & System.double P2z
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

P1x
X value of the line start point

P1y
Y value of the line start point

P1z
Z value of the line start point

P2x
X value of the line end point

P2y
Y value of the line end point

P2z
Z value of the line end point

Return Value

Newly created sketch line

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::CreateLine2](#).

Example

[Calculate Closest Distance Between Faces \(VBA\)](#)
[Calculate Closest Distance Between Model Components \(VBA\)](#)
[Create Section View at Specified Location \(VBA\)](#)
[Find Outside Edges of Face \(VBA\)](#)

Remarks

If a sketch is not active, then the line is not created and NULL is returned. You can check for an active sketch using [IModelDoc2::GetActiveSketch2](#) or [IModelDoc2::!GetActiveSketch2](#).

For COM applications, the underlying [ISketchLine](#) object can be obtained using QueryInterface on the [ISketchSegment](#) object returned. C++ Dispatch applications can define a new ISketchLine or ISketchSegment object that uses this Dispatch pointer. Visual Basic applications interpret the pointer for you automatically so you can use the returned object to call ISketchSegment or ISketchLine functions.

[IModelDoc2::SetAddToDB](#) and [IModelDoc2::SetDisplayWhenAdded](#) increase performance during entity creation by adding entities directly to the SOLIDWORKS database. IModelDoc2::SetAddToDB also avoids inferencing.

When this method is used with a drawing document, this method creates the line relative to the active drawing view, [IDrawingDoc::ActiveDrawingView](#) or [IDrawingDoc::!ActiveDrawingView](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::!CreateLine2 Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

CreateLinearSketchStepAndRepeat Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : CreateLinearSketchStepAndRepeat Method (IModelDoc2)

Obsolete. Superseded by [ISketchManager::CreateLinearSketchStepAndRepeat](#).

.NET Syntax

Visual Basic (Declaration)

```
Function CreateLinearSketchStepAndRepeat( _  
    ByVal NumX As System.Integer, _  
    ByVal NumY As System.Integer, _  
    ByVal SpacingX As System.Double, _  
    ByVal SpacingY As System.Double, _  
    ByVal AngleX As System.Double, _  
    ByVal AngleY As System.Double, _  
    ByVal DeleteInstances As System.String _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim NumX As System.Integer  
Dim NumY As System.Integer  
Dim SpacingX As System.Double  
Dim SpacingY As System.Double  
Dim AngleX As System.Double  
Dim AngleY As System.Double  
Dim DeleteInstances As System.String  
Dim value As System.Boolean
```

```
value = instance.CreateLinearSketchStepAndRepeat(NumX, NumY, SpacingX, SpacingY, AngleX, AngleY, DeleteInstances)
```

C#

```

System.bool CreateLinearSketchStepAndRepeat(
    System.int NumX,
    System.int NumY,
    System.double SpacingX,
    System.double SpacingY,
    System.double AngleX,
    System.double AngleY,
    System.string DeleteInstances
)

```

C++/CLI

```

System.bool CreateLinearSketchStepAndRepeat(
    & System.int NumX,
    & System.int NumY,
    & System.double SpacingX,
    & System.double SpacingY,
    & System.double AngleX,
    & System.double AngleY,
    & System.String^ DeleteInstances
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

NumX

Total number of instances along direction 1, including the seed

NumY

Total number of instances along direction 2, including the seed

SpacingX

Spacing between instances along direction 1

SpacingY

Spacing between instances along direction 2

AngleX

Relative to the X axis, the angle for direction 1

AngleY

Relative to the X axis, the angle for direction 2

DeleteInstances

Number of instances to delete passed as a string in the format: "(a) (b) (c) "

Return Value

True if the sketch pattern was created successfully, false otherwise

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::CreateLinearSketchStepAndRepeat](#).

See Also

[IModelDoc2 Interface](#)

[IModelDoc2 Members](#)

[IModelDoc2::CreateCircularSketchStepAndRepeat Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

CreateLineDB Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : CreateLineDB Method (IModelDoc2)

Obsolete. Superseded by [IModelDoc2::CreateLine2](#).

.NET Syntax

```
Function CreateLineDB( _
    ByVal Sx As System.Double, _
    ByVal Sy As System.Double, _
    ByVal Sz As System.Double, _
    ByVal Ex As System.Double, _
    ByVal Ey As System.Double, _
    ByVal Ez As System.Double _
) As System.Boolean
```

```
Dim instance As IModelDoc2
Dim Sx As System.Double
Dim Sy As System.Double
Dim Sz As System.Double
Dim Ex As System.Double
Dim Ey As System.Double
Dim Ez As System.Double
Dim value As System.Boolean

value = instance.CreateLineDB(Sx, Sy, Sz, Ex, Ey, Ez)
```

```
System.bool CreateLineDB(
    System.double Sx,
    System.double Sy,
    System.double Sz,
    System.double Ex,
    System.double Ey,
    System.double Ez
)
```

```
System.bool CreateLineDB(
    & System.double Sx,
    & System.double Sy,
    & System.double Sz,
    & System.double Ex,
    & System.double Ey,
    & System.double Ez
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Sx
Sy
Sz
Ex
Ey
Ez

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::CreateLineDB](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : CreateLineVB Method (IModelDoc2)

Obsolete. Superseded by [IModelDoc2::CreateLine2](#).

.NET Syntax

Visual Basic (Declaration)

```
Sub CreateLineVB( _  
    ByVal X1 As System.Double, _  
    ByVal Y1 As System.Double, _  
    ByVal Z1 As System.Double, _  
    ByVal X2 As System.Double, _  
    ByVal Y2 As System.Double, _  
    ByVal Z2 As System.Double _  
)
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim X1 As System.Double  
Dim Y1 As System.Double  
Dim Z1 As System.Double  
Dim X2 As System.Double  
Dim Y2 As System.Double  
Dim Z2 As System.Double  
  
instance.CreateLineVB(X1, Y1, Z1, X2, Y2, Z2)
```

C#

```
void CreateLineVB(  
    System.double X1,  
    System.double Y1,  
    System.double Z1,  
    System.double X2,  
    System.double Y2,  
    System.double Z2  
)
```

C++/CLI

```
void CreateLineVB(  
& System.double X1,  
& System.double Y1,  
& System.double Z1,  
& System.double X2,  
& System.double Y2,  
& System.double Z2  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

X1

Y1

Z1

X2

Y2

Z2

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::CreateLineVB](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

CreatePlaneAtAngle Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : CreatePlaneAtAngle Method (IModelDoc2)

Obsolete. Superseded by [IModelDoc2::CreatePlaneAtAngle3](#).

.NET Syntax

Visual Basic (Declaration)

```
Sub CreatePlaneAtAngle( _  
    ByVal Val As System.Double, _  
    ByVal FlipDir As System.Boolean _  
)
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim Val As System.Double  
Dim FlipDir As System.Boolean  
  
instance.CreatePlaneAtAngle(Val, FlipDir)
```

C#

```
void CreatePlaneAtAngle(  
    System.double Val,  
    System.bool FlipDir  
)
```

C++/CLI

```
void CreatePlaneAtAngle(  
& System.double Val,  
& System.bool FlipDir  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Val

FlipDir

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::CreatePlaneAtAngle](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

CreatePlaneAtAngle2 Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : CreatePlaneAtAngle2 Method (IModelDoc2)

Obsolete. Superseded by [IModelDoc2::CreatePlaneAtAngle3](#).

.NET Syntax

Visual Basic (Declaration)

```
Function CreatePlaneAtAngle2( _  
    ByVal Val As System.Double, _  
    ByVal FlipDir As System.Boolean _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim Val As System.Double  
Dim FlipDir As System.Boolean  
Dim value As System.Object  
  
value = instance.CreatePlaneAtAngle2(Val, FlipDir)
```

C#

```
System.object CreatePlaneAtAngle2(  
    System.double Val,  
    System.bool FlipDir  
)
```

C++/CLI

```
System.Object^ CreatePlaneAtAngle2(  
& System.double Val,  
& System.bool FlipDir  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Val

FlipDir

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::CreatePlaneAtAngle2](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

CreatePlaneAtAngle3 Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : CreatePlaneAtAngle3 Method (IModelDoc2)

Obsolete. Superseded by [IFeatureManager::InsertRefPlane](#).

.NET Syntax

Visual Basic (Declaration)

```
Function CreatePlaneAtAngle3( _  
    ByVal Val As System.Double, _  
    ByVal FlipDir As System.Boolean, _  
    ByVal AutoSize As System.Boolean _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim Val As System.Double  
Dim FlipDir As System.Boolean
```

```
Dim AutoSize As System.Boolean
Dim value As System.Object

value = instance.CreatePlaneAtAngle3(Val, FlipDir, AutoSize)
```

C#

```
System.Object CreatePlaneAtAngle3(
    System.double Val,
    System.bool FlipDir,
    System.bool AutoSize
)
```

C++/CLI

```
System.Object^ CreatePlaneAtAngle3(
    & System.double Val,
    & System.bool FlipDir,
    & System.bool AutoSize
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Val

Angle from the selected plane in radians

FlipDir

True to flip the plane the other way for first direction, false to not

AutoSize

True to automatically size the plane, false to not

Return Value

Newly created [reference plane](#)

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::CreatePlaneAtAngle3](#).

Remarks

This method uses the current document setting for displaying the reference plane. If display of reference planes is disabled, then you do not see the reference plane on the screen. If display of reference planes is enabled, then you see it. [IModelDocExtension::GetUserPreferenceToggle](#) swDisplayPlanes and [IModelDocExtension::SetUserPreferenceToggle](#) swDisplayPlanes get or set this display preference.

This method does not select the reference plane after it is created. Objects that are selected before running this method are still selected when the method completes, not the newly created reference plane.

This method returns an IRefPlane object. This object can then be used for subsequent operations on the reference plane feature. Having a IRefPlane object may not be useful, except that it is a feature, which is an entity, so methods available on those objects are available. For an OLE user, those functions are directly accessible; for a COM user, those functions are available via QueryInterface. For example, to select the reference plane, use [IEntity::Select4](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::CreatePlaneAtOffset3 Method](#)
[IModelDoc2::CreatePlaneAtSurface3 Method](#)
[IModelDoc2::CreatePlaneFixed2 Method](#)
[IModelDoc2::CreatePlanePerCurveAndPassPoint3 Method](#)
[IModelDoc2::CreatePlaneThru3Points3 Method](#)
[IModelDoc2::CreatePlaneThruLineAndPt Method](#)
[IModelDoc2::CreatePlaneThruPtParallelToPlane Method](#)
[IModelDoc2::CreatePlaneAtAngle3 Method](#)
[IModelDoc2::CreatePlaneAtOffset3 Method](#)
[IModelDoc2::CreatePlaneAtSurface3 Method](#)
[IModelDoc2::CreatePlaneFixed2 Method](#)
[IModelDoc2::CreatePlanePerCurveAndPassPoint3 Method](#)
[IModelDoc2::CreatePlaneThru3Points3 Method](#)
[IModelDoc2::CreatePlaneThruLineAndPt Method](#)
[IModelDoc2::GetVisibilityOfConstructPlanes Method](#)
[IModelDoc2::ViewDispRefPlanes Method](#)

Availability

CreatePlaneAtOffset Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : CreatePlaneAtOffset Method (IModelDoc2)

Obsolete. Superseded by [IModelDoc2::CreatePlaneAtOffset3](#).

.NET Syntax

Visual Basic (Declaration)

```
Sub CreatePlaneAtOffset( _  
    ByVal Val As System.Double, _  
    ByVal FlipDir As System.Boolean _  
)
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim Val As System.Double  
Dim FlipDir As System.Boolean  
  
instance.CreatePlaneAtOffset(Val, FlipDir)
```

C#

```
void CreatePlaneAtOffset(  
    System.double Val,  
    System.bool FlipDir  
)
```

C++/CLI

```
void CreatePlaneAtOffset(  
& System.double Val,  
& System.bool FlipDir  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Val

FlipDir

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::CreatePlaneAtOffset](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

CreatePlaneAtOffset2 Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : CreatePlaneAtOffset2 Method (IModelDoc2)

Obsolete. Superseded by [IModelDoc2::CreatePlaneAtOffset3](#).

.NET Syntax

Visual Basic (Declaration)

```
Function CreatePlaneAtOffset2( _  
    ByVal Val As System.Double, _  
    ByVal FlipDir As System.Boolean _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim Val As System.Double  
Dim FlipDir As System.Boolean  
Dim value As System.Object  
  
value = instance.CreatePlaneAtOffset2(Val, FlipDir)
```

C#

```
System.Object CreatePlaneAtOffset2(  
    System.double Val,  
    System.bool FlipDir  
)
```

C++/CLI

```
System.Object^ CreatePlaneAtOffset2(  
& System.double Val,  
& System.bool FlipDir  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Val

FlipDir

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::CreatePlaneAtOffset2](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[GetVisibilityOfConstructPlanes Method](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

CreatePlaneAtOffset3 Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : CreatePlaneAtOffset3 Method (IModelDoc2)

Obsolete. Superseded by [IFeatureManager::InsertRefPlane](#).

.NET Syntax

Visual Basic (Declaration)

```
Function CreatePlaneAtOffset3( _  
    ByVal Val As System.Double, _  
    ByVal FlipDir As System.Boolean, _  
    ByVal AutoSize As System.Boolean _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim Val As System.Double  
Dim FlipDir As System.Boolean
```

```
Dim AutoSize As System.Boolean
Dim value As System.Object

value = instance.CreatePlaneAtOffset3(Val, FlipDir, AutoSize)
```

C#

```
System.Object CreatePlaneAtOffset3(
    System.double Val,
    System.bool FlipDir,
    System.bool AutoSize
)
```

C++/CLI

```
System.Object^ CreatePlaneAtOffset3(
    & System.double Val,
    & System.bool FlipDir,
    & System.bool AutoSize
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Val

Offset distance from selected plane in meters

FlipDir

True to flip the plane the other way for first direction, false to not

AutoSize

True to automatically size the the plane, false to not

Return Value

Newly created [reference plane](#)

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::CreatePlaneAtOffset3](#).

Remarks

This method uses the current document setting for displaying the reference plane. If display of reference planes is disabled, then you do not see the reference plane on the screen. If display of reference planes is enabled, then you see it. [IModelDocExtension::GetUserPreferenceToggle](#) swDisplayPlanes and [IModelDocExtension::SetUserPreferenceToggle](#) swDisplayPlanes get or set this display preference.

This method does not select the reference plane after it is created. Objects that are selected before running this method are still selected when the method completes, not the newly created reference plane.

This method returns an IRefPlane object. This object can then be used for subsequent operations on the reference plane feature. Having a IRefPlane object may not be useful, except that it is a feature, which is an entity, so methods available on those objects are available. For an OLE user, those functions are directly accessible; for a COM user, those functions are available via QueryInterface. For example, to select the reference plane, use [IEntity::Select4](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::CreatePlaneAtAngle3 Method](#)
[IModelDoc2::CreatePlaneAtSurface3 Method](#)
[IModelDoc2::CreatePlaneFixed2 Method](#)
[IModelDoc2::CreatePlanePerCurveAndPassPoint3 Method](#)
[IModelDoc2::CreatePlaneThru3Points3 Method](#)
[IModelDoc2::CreatePlaneThruLineAndPt Method](#)
[IModelDoc2::CreatePlaneThruPtParallelToPlane Method](#)
[IModelDoc2::ICreatePlaneAtAngle3 Method](#)
[IModelDoc2::ICreatePlaneAtOffset3 Method](#)
[IModelDoc2::ICreatePlaneAtSurface3 Method](#)
[IModelDoc2::ICreatePlaneFixed2 Method](#)
[IModelDoc2::ICreatePlanePerCurveAndPassPoint3 Method](#)
[IModelDoc2::ICreatePlaneThru3Points3 Method](#)
[IModelDoc2::ICreatePlaneThruLineAndPt Method](#)
[IModelDoc2::ICreatePlaneThruPtParallelToPlane Method](#)
[IModelDoc2::GetVisibilityOfConstructPlanes Method](#)
[IModelDoc2::ViewDispRefplanes Method](#)

Availability

CreatePlaneAtSurface Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : CreatePlaneAtSurface Method (IModelDoc2)

Obsolete. Superseded by [IModelDoc2::CreatePlaneAtSurface3](#).

.NET Syntax

Visual Basic (Declaration)

```

Sub CreatePlaneAtSurface( _
    ByVal InterIndex As System.Integer, _
    ByVal ProjOpt As System.Boolean, _
    ByVal ReverseDir As System.Boolean, _
    ByVal NormalPlane As System.Boolean, _
    ByVal Angle As System.Double _
)

```

Visual Basic (Usage)



```

Dim instance As IModelDoc2
Dim InterIndex As System.Integer
Dim ProjOpt As System.Boolean
Dim ReverseDir As System.Boolean
Dim NormalPlane As System.Boolean
Dim Angle As System.Double

instance.CreatePlaneAtSurface(InterIndex, ProjOpt, ReverseDir, NormalPlane, Angle)

```

C#

```

void CreatePlaneAtSurface(
    System.int InterIndex,
    System.bool ProjOpt,
    System.bool ReverseDir,
    System.bool NormalPlane,
    System.double Angle
)

```

C++/CLI

```

void CreatePlaneAtSurface(
    & System.int InterIndex,
    & System.bool ProjOpt,
    & System.bool ReverseDir,
    & System.bool NormalPlane,
    & System.double Angle
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

InterIndex

ProjOpt

ReverseDir

NormalPlane

Angle

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::CreatePlaneAtSurface](#).

See Also

CreatePlaneAtSurface2 Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : CreatePlaneAtSurface2 Method (IModelDoc2)

Obsolete. Superseded by [IModelDoc2::CreatePlaneAtSurface3](#).

.NET Syntax

Visual Basic (Declaration)

```
Function CreatePlaneAtSurface2( _  
    ByVal InterIndex As System.Integer, _  
    ByVal ProjOpt As System.Boolean, _  
    ByVal ReverseDir As System.Boolean, _  
    ByVal NormalPlane As System.Boolean, _  
    ByVal Angle As System.Double _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim InterIndex As System.Integer  
Dim ProjOpt As System.Boolean  
Dim ReverseDir As System.Boolean  
Dim NormalPlane As System.Boolean  
Dim Angle As System.Double  
Dim value As System.Object  
  
value = instance.CreatePlaneAtSurface2(InterIndex, ProjOpt, ReverseDir, NormalPlane, Angle)
```

C#

```
System.object CreatePlaneAtSurface2(  
    System.int InterIndex,  
    System.bool ProjOpt,  
    System.bool ReverseDir,  
    System.bool NormalPlane,  
    System.double Angle  
)
```

C++/CLI

```
System.Object^ CreatePlaneAtSurface2(  
& System.int InterIndex,  
& System.bool ProjOpt,  
& System.bool ReverseDir,  
& System.bool NormalPlane,  
& System.double Angle  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

InterIndex

ProjOpt

ReverseDir

NormalPlane

Angle

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::CreatePlaneAtSurface2](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

CreatePlaneAtSurface3 Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : CreatePlaneAtSurface3 Method (IModelDoc2)

Obsolete. Superseded by [IFeatureManager::InsertRefPlane](#).

.NET Syntax

Visual Basic (Declaration)

```
Function CreatePlaneAtSurface3( _  
    ByVal InterIndex As System.Short, _  
    ByVal ProjOpt As System.Boolean, _  
    ByVal ReverseDir As System.Boolean, _  
    ByVal NormalPlane As System.Boolean, _  
    ByVal Angle As System.Double, _  
    ByVal AutoSize As System.Boolean _  
    ) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim InterIndex As System.Short  
Dim ProjOpt As System.Boolean  
Dim ReverseDir As System.Boolean  
Dim NormalPlane As System.Boolean  
Dim Angle As System.Double  
Dim AutoSize As System.Boolean  
Dim value As System.Object  
  
value = instance.CreatePlaneAtSurface3(InterIndex, ProjOpt, ReverseDir, NormalPlane, Angle, AutoSize)
```

C#

```
System.object CreatePlaneAtSurface3(  
    System.short InterIndex,  
    System.bool ProjOpt,  
    System.bool ReverseDir,  
    System.bool NormalPlane,  
    System.double Angle,  
    System.bool AutoSize  
)
```

C++/CLI

```
System.Object^ CreatePlaneAtSurface3(  
& System.short InterIndex,  
& System.bool ProjOpt,  
& System.bool ReverseDir,  
& System.bool NormalPlane,  
& System.double Angle,  
& System.bool AutoSize  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

InterIndex

- Multiple intersections - other solutions may exist
- a surface, plane, and edge - the intersection index is the intersection point to use when there are multiple intersections; when the intersection index input is more than the number of intersection points, the index of the last intersection point found will be used

ProjOpt

True to project the plane along the sketch normal toward the reference surface, false to project the plane to the nearest location on the reference surface

ReverseDir

True to create the plane on the opposite side of the sketch plane, false to not

NormalPlane

True to find the plane normal to the surface for a conical surface, false to find the plane tangent to the surface

Angle

Value of the angular offset of the normal plane, relative to a chosen reference plane

AutoSize

True to automatically size the plane, false to not

Return Value

Newly created [reference plane](#)

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::CreatePlaneAtSurface3](#).

Remarks

This method uses the current document setting for displaying of the reference plane as it is created.

If display of reference planes is...	Then you ...
Enabled	See the reference plane on the screen as it is created
Disabled	Do not see the reference plane on the screen as it is created

The [IModelDocExtension::GetUserPreferenceToggle](#) and [IModelDocExtension::SetUserPreferenceToggle](#) methods, with `swDisplayPlanes`, get or set this display preference.

This method does not select the reference plane after it is created. Objects that are selected before running this method are still selected when the method completes, not the newly created reference plane.

This method returns a `RefPlane` object. You can use this object for further operations on the reference plane feature. Although just having a reference plane may not be useful, it is a feature, which is an entity, so methods available on those objects are available.

For this type of user...	Those functions are...
OLE	Directly accessible
COM	Available via use of a <code>QueryInterface</code>

For example, if the reference plane must be selected, use [IEntity::Select4](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::CreatePlaneAtAngle3 Method](#)
[IModelDoc2::CreatePlaneAtOffset3 Method](#)
[IModelDoc2::CreatePlaneFixed2 Method](#)
[IModelDoc2::CreatePlanePerCurveAndPassPoint3 Method](#)
[IModelDoc2::CreatePlaneThru3Points3 Method](#)
[IModelDoc2::CreatePlaneThruLineAndPt Method](#)
[IModelDoc2::CreatePlaneThruPtParallelToPlane Method](#)
[IModelDoc2::ICreatePlaneAtAngle3 Method](#)
[IModelDoc2::ICreatePlaneAtOffset3 Method](#)
[IModelDoc2::ICreatePlaneAtSurface3 Method](#)
[IModelDoc2::ICreatePlaneFixed2 Method](#)
[IModelDoc2::ICreatePlanePerCurveAndPassPoint3 Method](#)
[IModelDoc2::ICreatePlaneThru3Points3 Method](#)
[IModelDoc2::ICreatePlaneThruLineAndPt Method](#)
[IModelDoc2::ICreatePlaneThruPtParallelToPlane Method](#)
[IModelDoc2::GetVisibilityOfConstructPlanes Method](#)
[IModelDoc2::ViewDispRefplanes Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

CreatePlaneFixed Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : CreatePlaneFixed Method (IModelDoc2)

Obsolete. Superseded by [IModelDoc2::CreatePlaneFixed2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function CreatePlaneFixed( _  
    ByVal P1 As System.Object, _  
    ByVal P2 As System.Object, _  
    ByVal P3 As System.Object, _  
    ByVal UseGlobal As System.Boolean _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim P1 As System.Object  
Dim P2 As System.Object  
Dim P3 As System.Object  
Dim UseGlobal As System.Boolean  
Dim value As System.Boolean  
  
value = instance.CreatePlaneFixed(P1, P2, P3, UseGlobal)
```

C#

```
System.bool CreatePlaneFixed(  
    System.object P1,  
    System.object P2,  
    System.object P3,  
    System.bool UseGlobal  
)
```

C++/CLI

```
System.bool CreatePlaneFixed(  
& System.Object^ P1,  
& System.Object^ P2,  
& System.Object^ P3,  
& System.bool UseGlobal  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

P1

P2

P3

UseGlobal

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::CreatePlaneFixed](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)



Obsolete. Superseded by [IFeatureManager::InsertRefPlane](#).

.NET Syntax

Visual Basic (Declaration)

```
Function CreatePlaneFixed2( _  
    ByVal P1 As System.Object, _  
    ByVal P2 As System.Object, _  
    ByVal P3 As System.Object, _  
    ByVal UseGlobal As System.Boolean _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim P1 As System.Object  
Dim P2 As System.Object  
Dim P3 As System.Object  
Dim UseGlobal As System.Boolean  
Dim value As System.Object
```

```
value = instance.CreatePlaneFixed2(P1, P2, P3, UseGlobal)
```

C#

```
System.object CreatePlaneFixed2(  
    System.object P1,  
    System.object P2,  
    System.object P3,  
    System.bool UseGlobal  
)
```

C++/CLI

```
System.Object^ CreatePlaneFixed2(  
& System.Object^ P1,  
& System.Object^ P2,  
& System.Object^ P3,  
& System.bool UseGlobal  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

P1

Array of 3 doubles (x, y, z) in meters; this is the first of three model-space points used to define the plane orientation; it is also used as the origin for the plane coordinate system

P2

Array of 3 doubles (x, y, z) in meters; this is the second of three model-space points used to define the plane orientation; the planes X-axis is directed from P1 to P2 unless useGlobal is set to True

P3

Array of 3 doubles (x, y, z) in meters; this is the final model-space point used to define the plane orientation

UseGlobal

Controls x-axis orientation (see **Remarks**)

Return Value

Newly created [reference plane](#) or NULL if the creation of the plane failed

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::CreatePlaneFixed2](#).

Remarks

The resulting plane is not parametric.

The planes normal vector is calculated using the cross product of the vectors (P2 - P1) and (P3 - P1), respectively.

The x-axis of the planes' coordinate system are a vector from P1 to P2 or a vector projected from the x-axis of the global coordinate system onto the plane.

The useGlobal argument denotes whether to align the x-axis of the plane with global orientation.

If useGlobal...	Then x-axis of the...
True	global (model) coordinate system is projected onto the plane. That vector is used to determine the direction of the plane's x-axis. This does not reorient the plane. Instead, it rotates the plane coordinate system about P1 until the x-axis of the plane aligns with the projected vector. P1, P2, and P3 are still required because they define the plane.
false	plane is aligned based on your input points, P1 and P2.

This method creates the plane in the model that is currently being edited. This behavior is consistent with the other plane creation methods, but is different from the original IModelDoc2::CreatePlaneFixed, which created the plane in this model, regardless of whether a component was currently being edited.

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::CreatePlaneAtAngle3 Method](#)
[IModelDoc2::CreatePlaneAtOffset3 Method](#)
[IModelDoc2::CreatePlaneAtSurface3 Method](#)
[IModelDoc2::CreatePlanePerCurveAndPassPoint3 Method](#)
[IModelDoc2::CreatePlaneThru3Points3 Method](#)
[IModelDoc2::CreatePlaneThruLineAndPt Method](#)
[IModelDoc2::CreatePlaneThruPtParallelToPlane Method](#)
[IModelDoc2::ICreatePlaneAtAngle3 Method](#)
[IModelDoc2::ICreatePlaneAtOffset3 Method](#)
[IModelDoc2::ICreatePlaneAtSurface3 Method](#)
[IModelDoc2::ICreatePlanePerCurveAndPassPoint3 Method](#)
[IModelDoc2::ICreatePlaneThru3Points3 Method](#)
[IModelDoc2::ICreatePlaneThruLineAndPt Method](#)
[IModelDoc2::ICreatePlaneThruPtParallelToPlane Method](#)
[IModelDoc2::GetVisibilityOfConstructPlanes Method](#)
[IModelDoc2::ViewDispRefPlanes Method](#)

Availability

SOLIDWORKS 2001Plus SP1, Revision Number 10.1

SOLIDWORKS API Help

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CreatePlanePerCurveAndPassPoint Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : CreatePlanePerCurveAndPassPoint Method (IModelDoc2)

Obsolete. Superseded by [IModelDoc2::CreatePlanePerCurveAndPassPoint3](#).

.NET Syntax

Visual Basic (Declaration)

```
Sub CreatePlanePerCurveAndPassPoint( _  
    ByVal OrigAtCurve As System.Boolean _  
)
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim OrigAtCurve As System.Boolean  
  
instance.CreatePlanePerCurveAndPassPoint(OrigAtCurve)
```

C#

```
void CreatePlanePerCurveAndPassPoint(  
    System.bool OrigAtCurve  
)
```

C++/CLI

```
void CreatePlanePerCurveAndPassPoint(  
& System.bool OrigAtCurve  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

OrigAtCurve

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::CreatePlanePerCurveAndPassPoint](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

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CreatePlanePerCurveAndPassPoint2 Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : CreatePlanePerCurveAndPassPoint2 Method (IModelDoc2)

Obsolete. Superseded by [IModelDoc2::CreatePlanePerCurveAndPassPoint3](#).

.NET Syntax

Visual Basic (Declaration)

```
Function CreatePlanePerCurveAndPassPoint2( _  
    ByVal OrigAtCurve As System.Boolean _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim OrigAtCurve As System.Boolean  
Dim value As System.Object  
  
value = instance.CreatePlanePerCurveAndPassPoint2(OrigAtCurve)
```

C#

```
System.object CreatePlanePerCurveAndPassPoint2(  
    System.bool OrigAtCurve  
)
```

C++/CLI

```
System.Object^ CreatePlanePerCurveAndPassPoint2(  
& System.bool OrigAtCurve  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

OrigAtCurve

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::CreatePlanePerCurveAndPassPoint2](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)